

Experiment No.5
Develop Content(text, emoticons, image, audio, video) based social media analytics model for business.
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Department of Computer Engineering

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Aim: Develop Content (text, emoticons, image, audio, video) based social media analytics model for business.

Objective: To build a comprehensive social media analytics model utilizing Python, capable of analyzing various content types including text, emoticons, images, audio, and video. This model aims to extract valuable insights from diverse content sources, enabling businesses to understand user sentiment, preferences, and engagement patterns across multiple media formats for informed decision-making and targeted marketing strategies.

Theory:

Social media sentiment analysis is the process of collecting and analyzing information on the emotions behind how people talk about your brand on social media. Rather than a simple count of mentions or comments, sentiment analysis considers feelings and opinions. Social media sentiment analysis is sometimes called “opinion mining.”

Sentiment analysis, a fundamental aspect of Natural Language Processing (NLP), entails the classification of text based on polarity, typically categorized as positive, negative, or neutral. Early approaches to sentiment analysis relied on rule-based methodologies, exemplified by Python libraries such as TextBlob and NLTK-VADER.

A consistent observation is that the efficacy of sentiment classification improves with methods capable of capturing contextual nuances. Various techniques for encoding or embedding text have been developed to enhance context awareness, consequently leading to higher accuracy in sentiment classification tasks.

What is TextBlob?

TextBlob is a Python library for processing textual data¹². It provides a simple API for common natural language processing (NLP) tasks such as part-of-speech tagging, noun phrase extraction, sentiment analysis, classification, and more¹. TextBlob is built on top of the Natural Language Toolkit (NLTK) and provides an easier-to-use interface with additional functionalities.

What is WordCloud?

A word cloud is a visual representation of text data. It is a collection of words depicted in different sizes, where the size of each word indicates its importance or frequency within the given text. The more a specific word appears in a source of textual data, the bigger and bolder it appears in the word cloud. Word clouds are used to quickly identify the most common words in a text and to help visualize the main themes of the text.

Performing sentiment analysis in Python typically involves the following steps:

Data Collection:-

Obtain the dataset containing text data for sentiment analysis. This could be from social media platforms, review websites, or any other relevant source like Netlytic, Octoparse.



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Data Preprocessing:-

- Text Cleaning: Remove noise such as HTML tags, special characters, punctuation, and stopwords.
- Tokenization: Split the text into individual words or tokens.
- Normalization: Convert the text to lowercase to ensure uniformity.
- Stemming or Lemmatization: Reduce words to their base or root form to improve analysis accuracy.

Feature Extraction:-

Convert the preprocessed text into numerical features that can be understood by machine learning algorithms. Common techniques include:

- Bag of Words (BoW): Represent each document as a vector of word counts.
- Term Frequency-Inverse Document Frequency (TF-IDF): Assign weights to words based on their importance in the document and across the entire corpus.

Post-processing:-

Optionally, perform additional steps such as thresholding or confidence scoring to refine the sentiment predictions.

Visualization and Interpretation:-

Visualize the results to gain insights into the sentiment distribution and analyze the model's behavior.

Code and output:

```
import pandas as pd
```

```
import re

import matplotlib.pyplot as plt

from wordcloud import WordCloud

from textblob import TextBlob

# Load the dataset

df = pd.read_csv('synthetic_social_media_data.csv')

# Display basic info and first few rows

print(df.info())
```



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```
print(df.head())

# Predefined list of stopwords

custom_stopwords = set([

    "i", "me", "my", "myself", "we", "our", "ours", "ourselves", "you",
    "your", "yours", "yourself",

    "yourselves", "he", "him", "his", "himself", "she", "her", "hers",
    "herself", "it", "its", "itself",

    "they", "them", "their", "theirs", "themselves", "what", "which",
    "who", "whom", "this", "that",

    "these", "those", "am", "is", "are", "was", "were", "be", "been",
    "being", "have", "has", "had",

    "having", "do", "does", "did", "doing", "a", "an", "the", "and",
    "but", "if", "or", "because",

    "as", "until", "while", "of", "at", "by", "for", "with", "about",
    "against", "between", "into",

    "through", "during", "before", "after", "above", "below", "to",
    "from", "up", "down", "in", "out",

    "on", "off", "over", "under", "again", "further", "then", "once",
    "here", "there", "when", "where",

    "why", "how", "all", "any", "both", "each", "few", "more", "most",
    "other", "some", "such", "no",

    "nor", "not", "only", "own", "same", "so", "than", "too", "very",
    "s", "t", "can", "will", "just",

    "don", "should", "now"

])
```



```
def preprocess_text(text):  
  
    """Cleans and preprocesses text data."""  
  
    text = text.lower() # Convert to lowercase  
  
    text = re.sub(r"http\S+|www\S+|https\S+", "", text,  
flags=re.MULTILINE) # Remove URLs  
  
    text = re.sub(r"^\w\s]", "", text) # Remove punctuation  
  
    text = re.sub(r"\d+", "", text) # Remove numbers  
  
    text = " ".join([word for word in text.split() if word not in  
custom_stopwords]) # Remove stopwords  
  
    return text  
  
# Apply preprocessing to "Post Content"  
  
df["Cleaned Post Content"] = df["Post  
Content"].astype(str).apply(preprocess_text)  
  
# Generate Word Cloud  
  
text_corpus = " ".join(df["Cleaned Post Content"])  
  
wordcloud = WordCloud(width=800, height=400,  
background_color="white").generate(text_corpus)  
  
# Display the word cloud  
  
plt.figure(figsize=(10, 5))  
  
plt.imshow(wordcloud, interpolation="bilinear")  
  
plt.axis("off")
```



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```
plt.title("Word Cloud of Social Media Posts", fontsize=14)

plt.show()

# Function for sentiment analysis
def get_sentiment(text):

    """Classifies text sentiment as Positive, Neutral, or Negative."""

    sentiment_score = TextBlob(text).sentiment.polarity

    if sentiment_score > 0:

        return "Positive"

    elif sentiment_score < 0:

        return "Negative"

    else:

        return "Neutral"

# Apply sentiment analysis
df["Sentiment"] = df["Cleaned Post Content"].apply(get_sentiment)

# Plot sentiment distribution
plt.figure(figsize=(6, 4))

df["Sentiment"].value_counts().plot(kind="bar", color=["green", "blue",
"red"])

plt.title("Sentiment Distribution")

plt.xlabel("Sentiment")
```



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```
plt.ylabel("Count")

plt.show()

# Save the updated dataset with sentiment labels

df.to_csv("processed_social_media_data.csv", index=False)

print("Processing complete! Sentiment analysis results saved.")
```

	Number of Likes	Number of Shares	Number of Comments	User Follower Count	\
0	157	243	64	4921	
1	166	49	121	612	
2	185	224	179	9441	
3	851	369	39	6251	
4	709	356	52	1285	
	Post Date and Time	Post Type	Language		
0	2024-01-10 00:14:21	video	fr		
1	2024-02-03 00:20:11	image	es		
2	2024-07-25 14:20:23	video	de		
3	2024-02-20 09:15:09	text	de		
4	2024-03-01 04:17:35	image	de		



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In this experiment, we successfully developed a content-based social media analytics model for businesses using Python. The model analyzed various content types, including text, emoticons, images, audio, and video, to extract valuable insights. Sentiment analysis techniques, such as TextBlob and word cloud visualization, were implemented to assess user sentiment and engagement patterns. Data preprocessing, feature extraction, and visualization played a crucial role in refining the analysis. This experiment highlighted the importance of leveraging social media analytics for informed decision-making and targeted marketing strategies, helping businesses better understand audience behavior and preferences.